



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester-B. Tech Information Technology

Course Code & Title : CS3391 Object Oriented Programming

Name of the Faculty member (s) : Dr. K. Palraj, ASP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 1 / Constructor
- **Course Outcome** : CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen** : Think Pair Share
- **Justification:**

Think–Pair–Share (TPS) strategy to teach constructors in Java is an innovative practice that enhances student engagement and deepens conceptual understanding through active participation. In the "Think" phase, students individually reflect on the purpose and usage of constructors, promoting independent critical thinking. During the "Pair" phase, peer discussions help clarify misconceptions and reinforce learning by comparing different perspectives, such as constructor overloading and validation logic. Finally, in the "Share" phase, students articulate their understanding with real-world examples, fostering higher-order thinking and communication skills. This approach transforms a technical topic into a collaborative learning experience, making complex programming concepts more accessible and meaningful.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - Students will be able to analyze and articulate strategies used by businesses to build and sustain strong customer relationships, and evaluate their effectiveness based on customer satisfaction, loyalty, and value.
 - From the 64 students, 32 groups are formed and assigned the topic to each group.

Team	Student	Topic
1.	Surulivelan M	Definition of Constructor and its uses
	Mathan Raja M	
2.	Varshini M	Types of Constructors
	Varuna Shree	
3.	Abinaya N	Constructor Overloading
	Adshayaa V	

4.	Deepikaa G	Discuss the program to create a class Student with a default constructor that prints “Student object created
	Devika R	
5.	Dhanya Sankari M	Discuss the program with a parameterized constructor in Book class to initialize title and author
	Harini K	
6.	Iptika D	Discuss the program with a class Rectangle that uses a constructor to initialize length and width, and a method to calculate area
	Ishwaryalakshmi V	
7.	Iswarya	Discuss the program to create a class Circle with a constructor that initializes radius and finds the area and circumference
	Kaleeswari A K	
8.	Karthiga R	Discuss the program to demonstrate constructor overloading in a class Person (default, name only, name with age)
	Kaviya K	
9.	Kiruthika M	Discuss the program to show the difference between default constructor and parameterized constructor in a class Car
	Lakshmi S	
10.	Mareeswari E	Discuss the program where a constructor initializes an array of integers and a method prints them.
	Murugalakshmi V	
11.	Nishika R S	Discuss the program with a class Employee that has a constructor to initialize employee details and display them
	Pavithra P	
12.	Pooja Shree S	Discuss the program to create a class BankAccount with a constructor to initialize account number and balance, and a method to deposit money
	Priya Rabiya T	
13.	Ram Selva Lakshmi M	Discuss the program with a class Triangle where constructor takes three sides and a method calculates area using Heron’s formula
	Santhiya S	
14.	Saranya S	Discuss the program to create a class Laptop with a constructor that initializes brand, model, and price, then display details
	Shanmuga Devi T	
15.	Shri Vethisha S	Discuss the program where a constructor copies values from one object to another (copy constructor concept)
	Suba Mala M	
16.	Vishnupriya R	Discuss the program to create a class Teacher with a constructor to initialize name, subject, and experience, then display the teacher profile
	Yamunadevi K	
17.	Arudayar Arun D	Discuss the program with constructor overloading in class Box (initialize with side, length & breadth, or copy another box)
	Shriram S	
18.	Anto Fieo R	Discuss the program where constructor initializes a matrix and a method prints its transpose
	Raagul G	
19.	Arumugaperumal G	Discuss the program with a class Mobile where constructor initializes brand and price, and method checks if the price is affordable (< 15000).
	Sudhakar R	
20.	Danny G	Discuss the program to create a class Movie with constructor to initialize name, rating, and duration, and method to display details.
	Sadagopan G	
21.	Hari Eswar M	Discuss the program with a class LibraryBook where constructor initializes book details, and method issues the book to a student
	Maheswaran R	

22.	D Vishal Nanda	Discuss the program to create a class Course where constructor initializes course name and number of students, then method prints enrollment status
	Vikashbalaji S	
23.	Kaniraj K	Discuss the program with a class Invoice where constructor initializes item, quantity, and price, and method calculates total bill
	Niraipandian B	
24.	Karthick S	Discuss the program to design a class ExamResult with constructor to initialize marks of three subjects and method to calculate average
	Maharaja T	
25.	Karthik M	Discuss the program with constructor in class ElectricityBill to initialize units consumed and method to calculate bill.
	Sudharshan S M	
26.	Lokesh R	Discuss the program with class TravelPackage where constructor initializes destination, days, and cost, and method displays details
	Thiraviyakumar J M	
27.	Mathivanan M	Discuss the program where a constructor initializes time (hours, minutes, seconds) and method converts to total seconds
	Suresh R	
28.	Miruthun Akshay M	Discuss the program where a parent class Vehicle has a constructor and child class Bike calls it using super()
	Shiri Shabari Sudhan	
29.	Naveen Karthick M	Discuss the program with a class Polynomial where constructor initializes coefficients and method evaluates the polynomial for a given x.
	Vishal P	
30.	Rishiguru Jo	Discuss the program with a class ComplexNumber where constructor initializes real and imaginary parts, and method adds two complex numbers.
	Sabari Subramanian	
31.	Zainudeen S	Discuss the program to implement a StudentDatabase where constructor initializes an array of student objects.
	Sivakumar M	
32.	Vinoth M	Discuss the program with a class ECommerceOrder where constructor initializes product, quantity, price, and method generates order summary.
	Balanavin S	

- Each student individually reflects on the question. They write down their own definition of a constructor and possibly a short example. Encourage them to think about why constructors are needed and what role they play in object creation. Its shown in Fig.1 and Fig.2.

- Next, Students pair up to discuss their thoughts and experiences.

In pairs, they discuss their answers and examples.

Prompt questions:

- Have you both define the constructor the same way?
- In what ways do your examples differ?
- Can constructors be overloaded? How?
- What happens if you don't define any constructor in a class?
- Finally, invite Surulivelan M and Mathan Raja M pair to share their discussion points with the whole class.
 - Ask pair to write sample constructor code on the board. Then compares different implementations and corrects misconceptions. Highlight best practices like using this keyword, input validation in constructors, and constructor chaining.

- Images / Screenshot of the practice



Figure 1. Pair with neighbour and discuss answer



Figure 2. Surulivelan M and Mathan Raja M are shared the answer to entire class

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	3	2	3	3	3	1	1	3	3	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	3	2	3	3	3	1	1	3	3	2
Justification for correlation	Students will be able to develop java program to solve the various engineering problems by using basics mathematics, science and engineering principles	Students will be able to understand problems, identify, and formulate solutions using various OOP concepts such as encapsulation, inheritance, and polymorphism to address both simple and complex engineering challenges.	Students will be able to develop solutions for real-time decision-making problems by applying appropriate Java packages along with conditional, branching, and looping statements.	Students will be able to use latest Java Development Kit to provide solution of the various real time problems	Students will be able to practice using tools like code editors, compilers, and version control systems to support object-oriented program development	Students are involved in team activities splitting their tasks, such as class creation, leading to better collaboration and shared learning.	Students will be able to Discuss their code and present their ideas clear to others.	Students will be able to manage project timelines and plan development phases for class-based implementations.	Students could be able to learning OOP helps students build skills for future learning in programming and objects	Students could be able to applies logic and programming principles to model real-world problems using classes and objects	Students could be able to understanding and applying the concepts of classes and objects is foundational for developing applications in emerging technologies like Mobile apps, IoT systems and Data analytics tools	Students could be able to encourages clean, modular code design as per software development best practices using OOP.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The TPS format encouraged even quieter students to participate, making the learning process more inclusive and engaging.
- Students felt more confident in expressing their ideas, as the pair discussion phase gave them time to clarify thoughts before speaking in front of the class.
- Many students reported that discussing the concept with peers helped them better understand the role and usage of constructors, especially constructor overloading and differences from regular methods.
- Students appreciated the interactive nature of the session, saying it broke the monotony of traditional lectures and made learning Java more fun.

- ❖ ***Benefit of the practice:***

- Students gain a deeper understanding of constructors by thinking individually, discussing with peers, and hearing diverse perspectives during sharing.
- Students clarify doubts and reinforce concepts by discussing with classmates, which often leads to better retention.
- Every student has a chance to contribute at their own comfort level, promoting equal participation and reducing fear of public speaking.

- ❖ ***Challenges faced in implementation:***

- Some students dominate the discussion while others remain passive, making it difficult to ensure equal engagement in pairs and groups.
- Students with limited proficiency in the medium of instruction may struggle to express their understanding clearly during pair or share phases.
- Some students may be reluctant to engage in interactive methods if they are used to traditional lectures or feel shy discussing in front of peers.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/KP_CS3591_Think_Pair_Share.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/KP_CCW332_Think_Pair_Share.pdf

Signature of Faculty Member

HOD